

PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

Augmented Analytics in SAP Analytics Cloud:

A Study of Impact and User Adoption

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In

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Augmented Analytics in SAP Analytics Cloud: A Study of Impact and User Adoption**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Preeti Saini**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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Abstract

While modern enterprises rapidly transition from conventional business intelligence architectures toward the development of AI-powered analytics ecosystems, a primary challenge lies in validating the operational promises of "automated insights." This project report analyzes the empirical efficiency and organizational adoption rates of Augmented Analytics within the SAP Analytics Cloud (SAC) platform. Moving beyond basic benchmark reviews of the technology's performance, this study explores how native features—specifically Smart Discovery, Smart Insights, and Natural Language Querying—impact enterprise decision-making pipelines.

Using an explanatory sequential mixed-methods research design, a dual tracking approach was implemented. First, a quantitative performance trial was conducted to evaluate execution speeds by processing an enterprise procurement dataset through parallel manual and automated workflows. Second, structural survey questionnaires were distributed to a sample of 63 active SAP professionals to map behavioral acceptance patterns.

The empirical findings demonstrate a substantial productivity multiplier, revealing that switching from manual data modeling to SAC's augmented tracks reduces time expenditure by up to 79% across primary analytical tasks, including an 84% reduction in identifying key data influencers. Furthermore, the automated engine achieved an outlier detection accuracy rate of 92%, outperforming manual reviews by 5 percentage points.

However, qualitative analysis highlights a critical behavioral "trust gap" stemming from the lack of algorithmic visibility. While users highly accept the platform as an analytical companion (scoring 4.4/5), they strongly resist relying on it as an autonomous decision-maker (scoring 2.6/5). The study concludes that technological advancement alone does not guarantee successful enterprise deployment; instead, actual system usage is heavily governed by data quality frameworks ($\rho = 0.73$), structured user training, and the integration of Explainable AI (XAI) features onto the enterprise roadmap to establish a transparent "cognitive partnership" between human analysts and machine models.

Keywords: Augmented Analytics, SAP Analytics Cloud, Smart Discovery, Time-to-Insight, Trust Gap, Explainable AI, Data Governance, Business Intelligence, SAP S/4HANA.

1. Introduction

1.1 Background and Evolution of BI

With the rapid evolution of digital business models, data has shifted from being a supplementary operational element to the primary driver of organizational growth and competitive advantage. However, as enterprise architectures expand, organizations encounter a complex "volume paradigm" generating data in massive quantities while struggling to efficiently translate it into actionable intelligence for real-time decision-making.

Traditional Business Intelligence (BI) systems were fundamentally built for an era when enterprise data volumes were manageable and reporting structures were highly centralized. These legacy solutions relied entirely on specialized data professionals to manually ingest, cleanse, slice, and interpret information, resulting in long analytical queues and delayed reporting cycles. In today's volatile market conditions, these static dashboards and retrospective structured reports are no longer sufficient to fulfill dynamic business requirements.

1.2 Core Technologies in Augmented Analytics

To address the limitations of traditional frameworks, Augmented Analytics has emerged as an essential evolutionary step, blending artificial intelligence (AI) and machine learning (ML) algorithms directly into business intelligence environments. The global augmented analytics market has demonstrated explosive scaling, expanding from an estimated valuation of USD 15.5 billion in 2024 to a projected USD 80.4 billion by 2033.

At its core, this technical revolution is underpinned by three foundational pillars:

- **Machine Learning (ML):** Automatically automates complex data preparation, profiles information, establishes hidden relationships, and constructs predictive models without human-written algorithms.
- **Natural Language Processing (NLP):** Empowers non-technical business professionals to interact with corporate data layers using everyday text or voice queries, removing the technical barrier of SQL or specialized query languages.
- **Natural Language Generation (NLG):** Automatically translates complex statistical variations, anomalies, and algorithmic insights into clear, plain-language text narratives, accelerating comprehension across corporate hierarchies.
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By combining these technologies, companies achieve the "democratization of data," enabling everyday users to navigate vast analytical ecosystems with minimal manual intervention.

1.3 Overview of SAP Analytics Cloud (SAC)

SAP Analytics Cloud (SAC) sits at the forefront of this transformation by maturely embedding unified planning, business intelligence, and AI-driven capabilities within a single enterprise cloud solution. Rather than forcing analysts to export corporate information to third-party tools, SAC provides an automated, streamlined workflow from initial raw insight to predictive business execution.

The platform achieves this through an advanced suite of smart features including Smart Discovery, Smart Insights, Smart Predict, and its conversational interfaces like Search to Insight and the generative AI copilot, Joule. By infusing generative AI into standard analytical processes, SAC enables modern enterprises to extract direct value from their core transactional backbones, such as SAP S/4HANA, leading to faster, automated, and more secure strategic data positioning.

2. Methodology

2.1 Research Design and Sequential Strategy

This study implements a mixed-methods research design, seamlessly merging quantitative performance evaluations with qualitative user-behavior tracking to form a comprehensive analysis of augmented analytics adoption barriers.

The structural blueprint utilizes an explanatory sequential strategy:

Phase 1: Quantitative Data Collection

Phase 2: Qualitative Data Exploration

Phase 3: Integrated Analysis & Synthesis

First, empirical performance timelines and data accuracy metrics were captured using task-based benchmarks. Second, structural survey questionnaires captured the behavioral patterns of active enterprise professionals. The resulting quantitative outputs—such as specific efficiency improvements—were then cross-referenced against qualitative thematic interviews to identify the root causes behind organizational trust dynamics and systemic adoption bottlenecks.

2.2 Research Questions and Objectives

The trajectory of this research is governed by four distinct Research Questions (RQs) designed to pinpoint technical value and user friction:

- **RQ1:** In what way does the use of augmented analytics functionalities in SAP Analytics Cloud save time compared to manual data modeling approaches?
- **RQ2:** How well do the insights produced using AI in SAC tools such as Smart Discovery and Smart Insights match the reality of business in corporate databases?
- **RQ3:** What is the main determiner, such as trust, user-friendliness, data quality, and organizational readiness, of the adoption of augmented analytics capabilities in SAP Analytics Cloud?

- **RQ4:** What is the perception of functional consultants and business analysts in relation to the validity of forecasts created through SAC's forecasting module?

2.3 Sampling and Data Collection Instruments

Data collection was executed using two primary investigative instruments:

1. Controlled Comparative Analysis

A rigorous performance trial was conducted by running the same enterprise procurement dataset through parallel workflows: the automated SAC augmented analytics track and a standard manual data-modeling track. Both pathways were tasked with identical analytical assignments: data profiling, outlier detection, trend profiling, and forward-looking predictions. Task timelines and discovery accuracy rates were measured against empirical baselines.

2. Structured Survey Questionnaire

A targeted survey was digitally distributed over a four-week window to a purposive sample of 63 active SAP Analytics Cloud professionals. The demographic breakdown comprised 68% SAP Functional Consultants and 22% Enterprise Business Analysts operating within large (52%) and medium-sized corporations. The qualifying threshold required a minimum of six months of direct, hands-on experience with SAC's smart feature ecosystem. The questionnaire was divided into five distinct domains: Demographics, Feature Utilization Patterns, Numerical Efficiency Metrics, Algorithmic Confidence Levels (anchored to the Technology Acceptance Model), and Open-Ended Feedback.

To secure statistical reliability, internal scale consistencies were strictly verified via Cronbach's Alpha, enforcing a verification floor of 0.70. Survey variables were grounded in Fred Davis's traditional Technology Acceptance Model (TAM), modified specifically to incorporate variables for output transparency, algorithmic opacity, and data governance controls.

3. Tools and Technologies

3.1 Smart Discovery and Smart Insights

- **Smart Discovery:** This automated machine-learning architecture is engineered to automatically parse underlying datasets and map out mathematically sound relationships without manual modeling. Upon execution, it dynamically constructs regression models for target measures and classification models for selected dimensions, evaluating and presenting the single most robust model based on accuracy and simplicity. The output is structured across a specialized four-page workspace: Overview, Key Influencers (variables exerting the most significant mathematical pull on the target metric), Unusual Values (anomalies), and a dynamic Simulation canvas.
- **Smart Insights:** Designed for rapid root-cause exploration, this tool provides instant statistical breakdowns of key points within visual charts. By selecting a specific data point or an unexpected cost spike, users trigger machine-learning sequences that instantly surface the primary contributing factors behind that specific variance, removing the need to manually build pivot tables or slice the data.

3.2 Search to Insight and Just Ask

- **Search to Insight:** Operating as SAC's foundation-level conversational interface, this feature utilizes localized natural language processing through an accessible, search-bar-driven interface. It interprets standard English phrasing to quickly query data models, returning instant, automated charts and visualizations bolstered by a context-aware autocomplete engine.
- **Just Ask:** Launched in Q1 2024, this represents SAC's next-generation conversational layer driven by generative AI. It expands far beyond basic search strings to parse highly complex, multi-variable queries. It automatically infers implicit parameters—such as contextual date filters or relative terms—and generates multi-faceted visualizations directly from natural language requests.

3.3 Smart Predict and Joule Copilot

- **Smart Predict:** This predictive engine democratizes advanced data science by enabling business personnel to compile high-fidelity forecasts without deep statistical expertise. The module handles underlying time-series calculations, trend extractions, and classification logic automatically, allowing advanced data scientists to stay focused on custom architectural programming.
- **Joule Copilot:** Seamlessly integrated into SAC's "Just Ask" engine, Joule serves as an advanced generative AI assistant across the broader SAP ecosystem. It blends transactional queries with conversational synthesis, generating real-time explanations, correlation matrices, and situational projections. Rather than presenting flat reporting blocks, Joule provides deep contextual insights, such as revealing how an supplier's lead time directly creates downstream budget exceptions.

4. Implementation

4.1 Data Ingestion and Workflow Configuration

The implementation framework was executed in a sequence of four distinct phases to map data from raw ingestion to full user evaluation:

Phase 1: Ingestion (Procurement Dataset)

Phase 2: Activation (Smart Suite & Joule)

Phase 3: Execution (Augmented Workflows)

Phase 4: Survey (User Evaluation)

- **Phase 1: Data Ingestion and Preparation:** A standard enterprise procurement dataset containing details on historical purchase orders, vendor cycles, lead times, and financial lines was uploaded into the SAP Analytics Cloud environment via an Excel repository. Ingestion cleaning was handled using native SAC data-profiling tools to resolve missing elements, normalize fields, and remove duplicate entries. This ensured that subsequent analytical deviations could be confidently attributed to algorithmic performance rather than data defects.
- **Phase 2: Feature Activation and Configuration:** Because SAC's smart capabilities are native to the tenant architecture, no external software installations were required. Operations required assigning relevant user permissions and tenant-level feature activations. Smart Discovery, Smart Insights, Just Ask, Smart Predict, and the Joule copilot layer were mapped directly to the cleaned procurement data model.
- **Phase 3: Execution of Augmented Analytics Workflows:** The analytical pipeline was processed in a structured sequence mimicking an enterprise analyst's exploratory routine:
 1. *Exploratory Parsing:* Smart Discovery was executed by assigning "Total Cost" as the primary target measure to automatically map key cost influencers and anomalous data records.
 2. *Root-Cause Tracking:* Smart Insights was used to query specific outlier vendors and sudden cost spikes.
 3. *Conversational Inquiries:* The "Just Ask" interface was queried with unstructured questions like "*What vendor type recorded the largest average cost overrun in Q3?*" to evaluate its natural language accuracy.
 4. *Predictive Projections:* Time-series predictive configurations were run in Smart Predict to map out expenditure lines for the upcoming two quarters.
- **Phase 4: User Interaction and Survey Execution:** Following the workflow testing, the 63 participating SAP professionals interacted with the system outputs. Over a four-week period, users evaluated these features using the structured TAM-based digital questionnaire to capture qualitative and quantitative feedback regarding platform usability and transparency.

4.2 Comparative Traditional Manual Workflow Setup

To establish a clear comparative baseline, the same procurement dataset was processed through a traditional, manual business intelligence track. This manual workflow required building custom calculated fields within standard data models, designing cross-dimensional pivot structures, mapping trend charts, and plotting regression formulas by hand.

Furthermore, the predictive forecasting step required building statistical time-series approximations within desktop spreadsheet software like Excel to match SAC's automated engine. Active processing durations, the number of distinct insights discovered, and the exact accuracy of anomaly captures were recorded and compared against the automated augmented track.

5. Results

5.1 Quantifying the Efficiency Dividend

The comparative analysis proved that SAC's analysis proved that SAC's augmented analytics capabilities serve as an immediate augmented analytics capabilities productivity multiplier for enterprise data management. Across the five foundational analytical tasks, switching from traditional manual setups to automated augmented workflows yielded an overall time reduction of up to 79%.

Analytical Task	Manual Duration (Mins)	Augmented Duration (Mins)	Absolute Time Reduction
Data Profiling & Summary Stats	18	4	78%
Outlier Detection	25	6	76%
Key Influencer Identification	32	5	84%
Trend Identification Across Dimensions	28	7	75%
Predictive Forecast Generation	45	8	82%

This empirical time reduction was validated by survey respondents, who returned a high mean score of 4.2 out of 5 for perceived efficiency gains. Qualitative commentary emphasized that these time savings were most valuable when exploring massive, multi-tiered data structures that typically trigger operational "analysis paralysis".

Crucially, speed did not compromise analytical depth. SAC's pattern recognition engine earned an average rating of 4.1 out of 5 from the user cohort. In outlier identification, Smart Discovery accurately flagged 92% of data anomalies, outperforming manual analyst reviews which caught 87%. In addition, the Key Influencers module delivered immediate value by surfacing unexpected, non-linear relationships—such as connecting specific supplier lead times to downstream project budget overruns.

5.2 The "Trust Gap" and Black Box Dynamics

Despite strong technical performance, a significant behavioral "trust gap" remains a key barrier to widespread adoption. The research highlights a sharp division in how users perceive AI:

User Agreement Ratings (out of 5):

AI as an Analytical Companion: 4.4 / 5

AI as an Autonomous Decider: 2.6 / 5

While professionals highly value AI as an analytical companion to speed up work (4.4/5), they show strong resistance to allowing the platform to act as an autonomous decision-maker (2.6/5). This resistance stems directly from the classic "Black Box Problem". With a mean score of just 2.9 out of 5 for user comprehension of how SAC builds its recommendations, the lack of algorithmic visibility remains a critical barrier to enterprise-wide adoption. Analysts are hesitant to stake operational budgets on automated insights without clear, step-by-step mathematical reasoning.

6. Conclusion and Future Scope

6.1 Roadmap Recommendations for SAP Systems

This study confirms that augmented analytics in SAP Analytics Cloud has shifted from a futuristic concept to a powerful accelerator. While the system delivers a near-80% efficiency dividend, user trust, platform explainability, and human data literacy remain the final challenges to successful adoption.

To bridge this trust gap and drive wider enterprise deployment, the following adjustments are recommended for future SAP technology roadmaps and organizational frameworks:

- **Native Integration of Explainable AI (XAI):** SAP should incorporate native XAI tools directly into its Smart Discovery and Smart Predict tracks. This includes adding clear feature-importance metrics, algorithmic confidence percentages, and exportable mathematical logs so analysts can audit and justify automated recommendations.
- **Enforcing Data Governance First:** Organizations must mature their data governance structures before deploying advanced analytics engines. Because data quality and AI precision share a strong correlation ($\rho = 0.73$), clean and consistent data models remain essential to getting reliable results from automated analytics.
- **Transitioning Training Frameworks:** User enablement strategies must move away from simple UI click-through training. Instead, training programs should focus on data literacy and conceptual foundations, empowering analysts to critically evaluate, challenge, and manage AI outputs within a true "cognitive partnership".

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